Electrical power and energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate electrical transfer rate and energy use.

SUCCESS CRITERIA - I can...

1. Use P=IV with watts, amps and volts.
2. Use E=Pt with seconds for joules.
3. Distinguish power from total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two equal **10 Ω** resistors are connected in parallel. **Find equivalent resistance**, then compare with a series connection of the same pair.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $P = IV$; $E = P\Delta t$; $P = I^2R = V^2/R$.
Constant power for $E = P\Delta t$; last equalities apply to an ohmic resistor.

Worked reasoning: A device at **6 V** drawing **2 A** receives power VI=**12 W**. Over **5 s** it receives **60 J**. Power is an energy-transfer rate.

Your turn: Draw an input-energy arrow and **label voltage**, current, power and elapsed time with distinct units.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two motors use the same energy but one takes half the time. **Compare average power**.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

04 Apply the physics

A **12 V** device draws a steady **0.75 A** for **4 minutes**.
1. **Calculate input power** and total electrical energy in joules.
2. **Label the diagram of the** objects you are studying and **show the time conversion**.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student says a **10 W** motor always uses more energy than a **5 W** motor. Supply a time comparison that disproves this.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A simplified shallow-water calculation uses v=√(gh). With g=**10**, h=**40 m** and travel distance **6 km**, find speed and travel time. **State a part of the** explanation that may not fit **what happens**.
Label a wave sketch to **support the calculation or comparison**.



Time or position: choose and label

Marzano check - circle: ① ② ③

07 Exit check

A **6 Ω** resistor carries **2 A**. **Find its power** using two equivalent equations.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **12 V** motor draws **1.5 A** for **40 s** and delivers **432 J** of useful mechanical energy. **Find electrical input energy**, average useful power and efficiency.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③